Levitation Lab

Technical Documentation

GT4Pebbles Project

May 15, 2026

This document describes the full physics model, numerical implementation, analytical tooling, user interface, and game design of the Levitation Lab browser simulation. It is intended for developers, researchers, and outreach coordinators who need to understand, extend, or verify the simulation.

Contents

1 Introduction

The **GT4Pebbles** (Ground Truth for Pebbles) project studies how planet formation begins in protoplanetary disks around young stars. A central experimental apparatus is a rotating drum in which dust particles are levitated for extended periods, allowing them to collide and grow through a sequence of stages: free particles → aggregates → pebbles → planetesimals.

The **Levitation Lab** is a browser-based interactive simulation that captures the core physics of this experiment. It is used both as an outreach tool—allowing members of the public to explore the experiment—and as a pedagogical tool for students and researchers. The simulation is built entirely in vanilla JavaScript and HTML5 Canvas, with no external physics engine.

This document covers:

- The underlying physical model and all governing equations.
- The numerical implementation, including timestep management and substep logic.
- Every user interface control, including button icons, with rationale for design choices.
- The game and challenge modes and their win/fail logic.
- The expert analytical panel and its visualisation tools.
- Audio, viewport, and URL parameter systems.

2 Coordinate System and Drum Geometry

2.1 Coordinate System

The simulation uses a right-handed coordinate system centred on the drum axis, with x pointing right and y pointing upward. Positions are measured in **drum units** (equivalent to centimetres in the physical experiment).

The drum has inner radius $R = 100\,\mathrm{du}$ and a hub of radius $R_{\mathrm{axis}} = 5\,\mathrm{du}$. The drum extrusion depth (into the screen) is $20\,\mathrm{du}$, but this dimension is not used in the 2-D simulation; it serves only as a conceptual reference.

Canvas rendering converts drum coordinates to screen pixels as:

$$x_{\mathrm{px}} = C_x + x \cdot S, \tag{1}$$

$$y_{\mathrm{px}} = C_y - y \cdot S, \tag{2}$$

where (C_x, C_y) is the screen-space drum centre and S is the pixel scale (pixels per drum unit), both computed by the layout engine from the window size.

2.2 Levitation Zone

The levitation zone (the *highlight zone* in the code) is a disc centred at $(H_{\mathrm{cx}}, H_{\mathrm{cy}}) = (50, 0)$ with radius $H_{\mathrm{r}} = 50\,\mathrm{du}$. This places the zone on the right side of the drum, tangent to both the drum wall and the axis. A particle is considered *levitated* when it has remained continuously inside the levitation zone for at least one full drum revolution.

3 Drum Rotation Dynamics

3.1 Angular Velocity Model

The drum has two angular velocity variables: the *target* velocity ω_{target} (set by the user via swipe or auto-omega) and the *actual* velocity ω.

The target decays exponentially each frame:

$$\omega_{\text{target}}(t + \Delta t) = \omega_{\text{target}}(t) \cdot e^{-\lambda_d \Delta t}, \tag{3}$$

with $\lambda_d = 1/30\,\text{s}^{-1}$ (i.e. a 30-second decay time constant). This models aerodynamic drag on the drum once the user stops spinning it. The decay can be disabled with the Ω **lock** button.

The actual velocity tracks the target with a slip rate $\lambda_s = 0.7\,\text{s}^{-1}$:

$$\omega(t + \Delta t) = \omega(t) + \left[\omega_{\text{target}}(t + \Delta t) - \omega(t)\right] \cdot \left(1 - e^{-\lambda_s \Delta t}\right). \tag{4}$$

3.2 User Input: Swipe

A pointer drag across the drum surface imparts angular momentum proportional to the tangential component of the drag velocity:

$$\Delta\omega_{\text{target}} = g_s \cdot \frac{\Delta s_{\text{tang}}}{r/R}, \tag{5}$$

where $g_s = 0.003\,(\text{rad}\,\text{s}^{-1})/\text{px}$ is the swipe gain, Δs_{tang} is the pointer displacement projected onto the tangential direction, and r is the distance of the midpoint from the axis. The denominator (r/R) normalises for swipes near the axis versus near the wall, giving consistent feel at all radii. The result is clamped to $|\omega_{\text{target}}| \leq \omega_{\max} = 2\pi \times 1.5\,\text{rad}\,\text{s}^{-1}$.

Design note. The physical experiment runs at approximately $1\,\text{RPM} \approx 0.105\,\text{rad}\,\text{s}^{-1}$. At that speed, the levitation feedback to the user would be far too slow for an interactive experience. The simulation maximum of $1.5\,\text{rev}\,\text{s}^{-1}$ is approximately 90 times faster, but preserves all the qualitative physics.

4 Particle Kinematics and Levitation

4.1 Equations of Motion

In the rotating frame, the gas co-rotates with the drum and exerts a drag force on the particle that brings its velocity to the local gas velocity minus its terminal settling velocity v_{t}. In the (non-rotating) lab frame, the particle's instantaneous velocity is:

$$\dot{x} = -\omega y, \tag{6}$$
$$\dot{y} = +\omega x - v_{\text{t}}. \tag{7}$$

Equation (6)–(7) give the gas velocity at the particle's position (a co-rotating body has $\dot{x} = -\omega y$, $\dot{y} = +\omega x$), reduced by v_{t} in the y-direction. The settling velocity v_{t} is always directed downward (the $-\hat{y}$ direction) and represents the particle's Stokes terminal velocity in quiescent gas.

This is a *kinematic* model: accelerations are not integrated; the velocity is set to the equilibrium value at every substep. This is valid when the aerodynamic drag time is much shorter than the orbital period, which is the overdamped (high-Stokes-drag) limit appropriate for the small particles of interest.

4.2 Circular Orbits

Setting $\ddot{x} = \ddot{y} = 0$ in the Newtonian sense (the orbit is closed) gives the orbit centre:

$$\boxed{x_\mathrm{c} = \frac{v_\mathrm{t}}{\omega}, \qquad y_c = 0.} \tag{8}$$

A particle at (x_0, y_0) therefore orbits a circle of radius

$$r = \sqrt{(x_0 - x_\mathrm{c})^2 + y_0^2} \tag{9}$$

centred at $(x_\mathrm{c}, 0)$. The orbit traces out at angular rate ω (the same rate as the drum).

From eq. (8), heavier particles (larger v_t) orbit at larger x_c and therefore closer to the drum wall. For a particle to remain levitated, its entire orbit must stay within the levitation zone. Since the levitation zone is centred at $(H_\mathrm{cx}, 0) = (50, 0)$ with radius $H_\mathrm{r} = 50$, this corresponds roughly to $x_\mathrm{c} \approx 50$ and $r \lesssim 30$–$40\,\mathrm{du}$.

4.3 Terminal Velocity Distributions

Each injection event draws N_P particles with terminal velocities from one of three distributions:

4.3.1 Gaussian (default)

$$v_\mathrm{t}^{(i)} = \bar{v}_\mathrm{t} \cdot \big(1 + s \cdot \xi_i\big), \qquad \xi_i \sim \mathcal{N}(0, 1), \tag{10}$$

where $\bar{v}_\mathrm{t}$ is the mean terminal velocity and s is the relative spread parameter (SPREAD slider, $0 \leq s \leq 0.5$).

4.3.2 Bi-disperse

Two groups with terminal velocities $v_\mathrm{t}^{(1)}, v_\mathrm{t}^{(2)}$, fractional spreads s_1, s_2, and mixing ratio q (particles in group 1 per particle in group 2):

$$P(\text{group 1}) = \frac{q}{1+q}. \tag{11}$$

Within each group, the terminal velocity is Gaussian with the group's own $\bar{v}_\mathrm{t}{}^{(k)}$ and s_k.

4.3.3 Power-law

The differential size distribution $n(v) \propto v^q$ on $[v_{\min}, v_{\max}]$ is sampled by inverse-transform:

$$v_\mathrm{t} = \left[u\left(v_{\max}^{q+1} - v_{\min}^{q+1}\right) + v_{\min}^{q+1}\right]^{1/(q+1)}, \qquad u \sim \mathrm{Uniform}(0,1), \tag{12}$$

with the special case $q = -1$ (log-uniform):

$$v_\mathrm{t} = v_{\min} \cdot \left(\frac{v_{\max}}{v_{\min}}\right)^u. \tag{13}$$

In all cases, v_t is clamped from below at the *settle floor* $= 2\,\mathrm{cm\,s^{-1}}$.

4.4 Visual Size

The rendered radius of a particle scales as:

$$s_\mathrm{vis} = \sqrt{\frac{v_\mathrm{t}}{v_\mathrm{t}^\mathrm{ref}}}, \qquad v_\mathrm{t}^\mathrm{ref} = 30\,\mathrm{cm\,s^{-1}}, \tag{14}$$

clamped to $[s_{\min}, s_{\max}] = [0.55, 1.8]$ drum units, and further scaled down by a factor 0.1–0.7 at large particle counts ($N_P \geq 300$) to keep the display readable.

4.5 Injection and Release

Particles are released from a brass manifold above the drum at $y = 110\,\mathrm{du}$ (above the wall), spread uniformly in x across $[0, 100]\,\mathrm{du}$. They fall freely (no drag) until they cross the inner drum wall, at which point the kinematic drag model (6)–(7) activates. Injection events are evenly spaced over the injection interval Δt_inj; a puff animation accompanies the start of injection (one puff per nozzle, no further puffs regardless of particle count, to avoid GPU performance issues on mobile).

4.6 Wall Collision and Particle Loss

At each substep, if a non-stuck particle's distance from the axis satisfies $r \geq R_\mathrm{wall} = R - r_\mathrm{coll}$ (where $r_\mathrm{coll} = 0.5\,\mathrm{du}$ is the collision radius), the particle is *stuck*: it is fixed to the wall at the collision angle and rotates with the drum. Stuck particles are displayed in red, fade over 5 s, and then removed. The `lostCount` accumulator is incremented.

With **Nanocoating** active, wall contact is handled elastically instead: the radial component of the velocity is reflected, and the particle is clamped just inside the wall.

5 Automatic Drum Speed Control

5.1 Orbit-Centring Formula

The auto-omega function computes the drum speed that centres the range of orbit centres for the current v_t distribution inside the levitation zone.

From eq. (8), an orbit at v_t is centred at $x_\mathrm{c} = v_\mathrm{t}/\omega$. For a distribution spanning $[v_\mathrm{t}^\mathrm{min}, v_\mathrm{t}^\mathrm{max}]$, the orbit centres span $[v_\mathrm{t}^\mathrm{min}/\omega,\ v_\mathrm{t}^\mathrm{max}/\omega]$. Centring this range on $H_\mathrm{cx} = 50$ gives:

$$\boxed{\omega_\mathrm{auto} = \frac{v_\mathrm{t}^\mathrm{min} + v_\mathrm{t}^\mathrm{max}}{2H_\mathrm{cx}}.} \tag{15}$$

For the Gaussian distribution, $v_\mathrm{t}^\mathrm{min} = \bar{v}_\mathrm{t}(1-s)$, $v_\mathrm{t}^\mathrm{max} = \bar{v}_\mathrm{t}(1+s)$, giving $\omega_\mathrm{auto} = \bar{v}_\mathrm{t}/H_\mathrm{cx}$, independent of spread.

For bi-disperse and power-law distributions, the full envelope of the distribution is used. The result is clamped to $[0.02, \omega_\mathrm{max}]$.

5.2 Launch Mode

In *launch mode* (three-state auto-omega button, second state), the controller waits for particles to be injected, estimates the time of peak particle flux at $y = 0$, spins the drum up just before that time using a kernel-density estimate of the arrival time distribution, then transitions to live tracking. The optimal spin-up time is:

$$t_\mathrm{wait} = \hat{t}_\mathrm{peak} - \frac{1}{\lambda_s}, \tag{16}$$

where $\hat{t}_\mathrm{peak}$ is the KDE mode of arrival times $t_\mathrm{arr}^{(i)} = t_\mathrm{inj}^{(i)} + R_\mathrm{release}/v_\mathrm{t}^{(i)}$ and the subtracted term accounts for the drum slip lag.

6 Aggregate Formation

6.1 Formation Threshold and Trigger

Aggregate formation becomes possible when at least $N_\mathrm{lev}^\mathrm{min} = 30$ particles are simultaneously levitated and the velocity spread satisfies $s \geq s_\mathrm{min}^\mathrm{agg} = 0.20$.

Once the threshold is met, a counter accumulates in units of drum revolutions:

$$\mathcal{R}_\mathrm{agg} \mathrel{+}= \frac{\Delta t}{T}, \qquad T = \frac{2\pi}{\omega}. \tag{17}$$

The first aggregate forms after $\mathcal{R}_\mathrm{agg} \geq 3$ revolutions; each subsequent one after ≥ 1 revolution (scaled down at large N_P to maintain visual interest).

6.2 Merge Animation

When triggered, $n_\mathrm{merge} = 10$ levitated particles near a randomly chosen levitated particle are selected as the nearest cluster. They are animated along straight paths toward their centroid over a duration $\tau_\mathrm{merge} = 0.6\,\mathrm{s}$, then replaced by a single aggregate.

6.3 Aggregate Terminal Velocity and Stokes Kick

The aggregate's terminal velocity is the mass-weighted mean of its constituent monomers, scaled by a Stokes-number factor:

$$v_\mathrm{t}^\mathrm{agg} = \bar{v}_\mathrm{t} \cdot f_\mathrm{Stokes}, \qquad f_\mathrm{Stokes} = 1.2 \text{ (when Stokes kick is on).} \tag{18}$$

The factor $f_{\text{Stokes}} > 1$ reflects two physical effects acting in the same direction:

1. **Increased inertia**: a compact aggregate has a higher Stokes number (ratio of aerodynamic response time to orbital period) than an individual monomer, so it settles faster in the rotating gas frame.
2. **Cross-section shadowing**: the aggregate's projected area is smaller than the sum of its constituent areas (for compact, roughly spherical aggregates), reducing drag per unit mass.

The effect is that aggregates orbit at a slightly larger x_{c} than the monomer population from which they formed, enabling them to orbit closer to the wall and reducing the risk of further levitation loss.

When **Stokes kick** is off, $f_{\text{Stokes}} = 1$ and the aggregate orbits exactly at the mean monomer orbit centre.

6.4 Aggregate Orbital Mechanics

Aggregates follow the same kinematic orbit equations (6)–(7) as monomers, but with their own $v_{\text{t}}^{\text{agg}}$. Additionally, when **aggregate growth mode** is active, turbulent perturbations are applied each substep:

$$\delta v_x = \sigma_B \xi_x, \tag{19}$$

$$\delta v_y = \sigma_B \xi_y, \tag{20}$$

where $\xi_{x,y} \sim \mathcal{N}(0,1)$ and $\sigma_B = 2\,\text{cm}\,\text{s}^{-1}$. A restoring force returns the aggregate toward its natural orbit centre $(x_{\text{c}}, 0)$:

$$\dot{v}_x^{\text{restore}} = k_R(x_{\text{c}} - x), \tag{21}$$

$$\dot{v}_y^{\text{restore}} = k_R(0 - y), \tag{22}$$

with spring constant $k_R = 0.5\,\text{s}^{-1}$. This is a Langevin-type equation that produces a random walk around the orbit centre with a characteristic diffusion length set by σ_B/k_R.

The perturbations cause aggregate orbits to drift, enabling close approaches with other aggregates.

6.5 Aggregate–Aggregate Growth Collisions

When growth mode is active (orange state), aggregate pairs are checked for overlap each frame. When two aggregates touch ($d < r_i + r_j$), the smaller merges into the larger over $\tau_{\text{grow}} = 0.6\,\text{s}$:

$$n_{\text{new}} = n_i + n_j, \tag{23}$$

$$v_{\text{t}}^{\text{new}} = \frac{n_i v_{\text{t}}^{(i)} + n_j v_{\text{t}}^{(j)}}{n_i + n_j} \cdot f_{\text{grow}}, \tag{24}$$

where $f_{\text{grow}} = 1.05$ per growth step (when Stokes kick is on). The visual radius grows as:

$$r_{\text{agg}} = r_{\text{coll}} \cdot s_{\text{vis}} \cdot n_{\text{mult}} \cdot \sqrt{n/10}, \tag{25}$$

with $n_{\text{mult}} = 10$ (the aggregate size multiplier) and n the current monomer count.

An aggregate reaching $n \geq 100$ monomers automatically becomes a **pebble** (golden ball).

6.6 Wall Collision of Aggregates

An aggregate reaching the drum wall fragments into 10 stuck monomers spread around the impact angle, and the aggregate count is decremented. With **Nanocoating** active, the aggregate is instead elastically reflected.

7 Pebble (Golden Ball) Formation

7.1 Formation Threshold

After $n_{\text{agg}}^{\text{crit}} = 7$ aggregates are simultaneously levitated for a hold time of $H_{\text{pebble}} = 3$ revolutions (first pebble) or 3 revolutions (subsequent), a pebble merge is triggered. The velocity spread condition $s \geq s_{\text{min}}^{\text{pebble}} = 0.30$ must also be satisfied.

The aggregate targets are chosen by a K-nearest-neighbours search ($K = 3$) to find the densest cluster, producing visually coherent merges. The $n_{\text{agg}}^{\text{crit}} = 7$ chosen aggregates are animated toward their centroid over $\tau_{\text{pebble}} = 0.6\,\text{s}$, and the aggregate count is reduced by 7 on completion.

7.2 Golden Ball Physics

Golden balls obey Newtonian dynamics, not the kinematic drag model:

$$\ddot{x} = 0, \tag{26}$$

$$\ddot{y} = -g = -200\,\text{du}\,\text{s}^{-2}. \tag{27}$$

They bounce off the drum wall with restitution coefficient $e_w = 0.85$ (normal) and tangential friction $\mu_w = 0.92$:

$$v_n^+ = -e_w v_n^-, \tag{28}$$

$$v_t^+ = \mu_w v_t^-, \tag{29}$$

where superscripts $\pm$ denote post- and pre-collision components.

Ball–ball collisions are handled with coefficient $e_b = 0.90$:

$$J = \frac{(1 + e_b)\,v_{\text{rel,n}}}{2}, \qquad \Delta\mathbf{v}_{A,B} = \mp J\,\hat{n}. \tag{30}$$

A rotating **bump** fixed to the drum wall (at the same angular position as the collection tray slot) collides with balls as the drum rotates, adding energy to keep them bouncing.

Wall contact drag interpolates the ball's tangential velocity toward the drum surface velocity with time constant $\lambda_c = 0.7\,\text{s}^{-1}$.

8 Globe (Planet) Formation

8.1 Formation Threshold

When $n_{\text{ball}}^{\text{crit}} = 3$ golden balls are simultaneously present, they merge into a **globe** over $\tau_{\text{globe}} = 1.2\,\text{s}$. Up to 7 globes can exist simultaneously.

8.2 Globe Hover Physics

Each globe is attracted toward the hover point $(0, -50)$ (drum units) by a spring:

$$\dot{v}_x \mathrel{+}= 0.01 \cdot 60 \cdot (0 - x)\,\Delta t, \tag{31}$$
$$\dot{v}_y \mathrel{+}= 0.01 \cdot 60 \cdot (-50 - y)\,\Delta t, \tag{32}$$

and repelled from all other globes by a $1/r^2$ force with constant $F_{\text{rep}} = 15000\,\text{du}^3\,\text{s}^{-2}$, with an additional linear hard-contact push when $d < 2.2R_{\text{globe}}$.

Velocity is damped at $\exp(-9.74\,\Delta t)$ per frame, equivalent to $\approx 0.85^{1/\Delta t}$ at 60 fps.

9 Solar System Phase Machine

Once a second globe forms, the simulation enters a solar system mode and progresses through a finite state machine: `showing` → `transitioning` → `orbiting` → `spindown` → `final_move` → `final_view`.

9.1 Orbital Dynamics

Each globe is assigned an orbit with semi-major axis from the sequence $[11, 17, 24, 31, 37, 42, 50]$ du (scaled dynamically to fit all globes inside the drum). The orbital angular velocity follows a modified Keplerian law:

$$\omega_{\text{orbit}}(r) = \frac{\omega_0}{r^{0.75}}, \qquad \omega_0 \approx 3.006\,\text{rad}\,\text{s}^{-1}, \tag{33}$$

where $\omega_0 = (2\pi/8) \cdot 6^{0.75}$.

The exponent 0.75 is shallower than the true Keplerian value of 1.5. This is a deliberate design choice: the full Keplerian scaling would make inner planets orbit extremely fast relative to outer ones, which looks chaotic at the speed of the simulation. The 0.75 exponent gives a visually convincing impression of varying orbital periods while keeping all orbits visible at once.

The orbits are displayed as inclined ellipses with inclination $\cos i = 0.3$ (i.e. $i \approx 73°$), giving a sense of a tilted orbital plane viewed in 3-D perspective.

9.2 System Scale

As new globes are added, the system scale $\mathcal{S}$ is reduced so that all orbits fit within the drum:

$$\mathcal{S}_n = \frac{R_{\max}}{r_n + r_{\text{orb}}}, \tag{34}$$

where $R_{\max} = 80$ du is the maximum safe orbit radius, r_n is the outermost orbit radius for n globes, and $r_{\text{orb}} = 3$ du is the visual orbit size. Transitions between scales are eased with a cubic smoothstep over 3 s.

9.3 Voyager Probes

After the solar system reaches its final static state (all seven planets, drum stopped), two probe objects are launched sequentially (with a 2.4 s gap) from the innermost planet. Each probe visits all planets in order at speed $28\,\mathrm{du\,s^{-1}}$, then escapes radially to the drum edge. The probes leave luminous trails. When both probes have completed their escape, an end-of-game sheet is shown after a 2 s delay.

10 Numerical Implementation

10.1 Timestep and Substep Logic

The main loop runs at the browser's animation frame rate (≈ 60 fps on most displays). The wall-clock frame duration $\Delta t_{\mathrm{frame}}$ is capped at $\Delta t_{\max} = 0.033\,\mathrm{s}$ to prevent large jumps after tab switches or garbage collection pauses. A simulation speed multiplier $s_{\mathrm{sim}} \in [0.25, 4]$ is applied:

$$\Delta t = \min(\Delta t_{\mathrm{frame}}, \Delta t_{\max}) \cdot s_{\mathrm{sim}}. \tag{35}$$

Physics are advanced in substeps of size $h = \Delta t / n_{\mathrm{sub}}$, where:

$$n_{\mathrm{sub}} = \max\left(1, \left\lceil \frac{\Delta t}{\Delta t_{\mathrm{sub}}} \right\rceil\right), \qquad \Delta t_{\mathrm{sub}} = \frac{T_{\min}}{100} = \frac{2\pi/\omega_{\max}}{100}. \tag{36}$$

At $\omega_{\max} = 2\pi \times 1.5\,\mathrm{rad\,s^{-1}}$, $\Delta t_{\mathrm{sub}} \approx 6.7 \times 10^{-3}\,\mathrm{s}$, giving ≈ 10 substeps per frame at 60 fps. This ensures that even the fastest orbiting particles do not cross the collision margin in a single substep.

10.2 Slow-Motion During Merges

When the slow-motion flag is armed (), any frame during which a merge animation is active (aggregate, pebble, or globe) is advanced at 1/5 real speed:

$$\Delta t \to \Delta t/5 \quad \text{(when merging)}. \tag{37}$$

This does not affect the audio system, which runs on the Web Audio API and is independent of the simulation clock.

10.3 Stroboscopic Mode

When strobe mode is active (), the canvas is redrawn only once per complete drum revolution. The revolution counter is tracked as $\lfloor |\theta_{\mathrm{drum}}|/2\pi \rfloor$; when this advances, a redraw is triggered. Levitated particles appear stationary; drifting particles advance between flashes, revealing their motion.

10.4 Simulation Speed Gears

Five speed gears are available: $\{0.25\times, 0.5\times, 1\times, 2\times, 4\times\}$. The multiplier applies to Δt before substep splitting. The drum decay, injection timing, particle injection, and all physics scale with the simulation clock, so all processes are accelerated or decelerated uniformly.

10.5 Pause

Pressing `Space` pauses the simulation. All physics, drum rotation, and motor audio freeze; the canvas continues to render the frozen state.

11 Heatmap and Analytical Overlays

11.1 Heatmap Accumulation

The drum interior is divided into a 50×50 grid. Each substep, each live non-stuck particle is binned by position, and three quantities are accumulated per cell: n (count), Σv (speed sum), and Σv^2 (speed-squared sum).

At the end of each complete drum revolution, the following quantities are finalised for each cell:

$$\bar{v} = \frac{\Sigma v}{n}, \tag{38}$$

$$\sigma_v = \sqrt{\frac{\Sigma v^2}{n} - \bar{v}^2} \quad \text{(velocity dispersion)}, \tag{39}$$

$$\rho = \frac{n}{N_{\text{ticks}}} \quad \text{(mean particle density per frame)}, \tag{40}$$

$$\mathcal{C} = \rho \cdot \sigma_v^2 \quad \text{(collision proxy)}. \tag{41}$$

The collision proxy $\mathcal{C} = \rho\sigma_v^2$ is motivated by the kinetic theory expression for the collision rate between two particle populations: $R_{12} \propto n_1 n_2 \sigma_{\text{rel}}$, where $\sigma_{\text{rel}} \propto \sigma_v$ for particles with different v_{t}. The proxy therefore highlights regions where both population density and velocity dispersion are high — i.e. where aggregate formation is most likely.

Three display modes cycle through with the ◀ ▶ buttons:

1. **Velocity variance** σ_v — shows where particles have the largest spread in speed, indicating different orbital radii crossing.
2. **Particle density** ρ — shows where particles spend the most time.
3. **Collision proxy** $\mathcal{C} = \rho\sigma_v^2$ — combines the two, highlighting the most collision-active region.

11.2 Orbit Projections

The *selected orbits* mode draws, for a representative sample of particles:

- The orbit circle centred at $(x_{\text{c}}, 0) = (v_{\text{t}}/\omega, 0)$ with radius $r = \sqrt{(x - x_{\text{c}})^2 + y^2}$.
- A dot marking x_{c} on the x-axis.
- Green circles for levitated particles, red for non-levitated.

The sample is updated every 3 s and comprises: the 5 smallest-circumference orbits, up to 10 levitated particles spanning the full x_{c} range, and up to 10 non-levitated particles spanning their x_{c} range.

11.3 Aggregate Encounter Prediction

For each pair of live aggregates, the simulation analytically computes the *closest approach* position during the next orbit. Each aggregate follows a circular orbit:

$$x_i(\theta) = x_{\rm c}^{(i)} + R_i \cos(\phi_i + \theta), \tag{42}$$

$$y_i(\theta) = R_i \sin(\phi_i + \theta), \tag{43}$$

where θ is the orbital phase advance (both advance at the same rate ω). The squared separation is:

$$d^2(\theta) = C_0 + 2\,\Delta x_{\rm c}\,(P\cos\theta - Q\sin\theta), \tag{44}$$

where $P = R_i\cos\phi_i - R_j\cos\phi_j$, $Q = R_i\sin\phi_i - R_j\sin\phi_j$, $\Delta x_{\rm c} = x_{\rm c}^{(i)} - x_{\rm c}^{(j)}$. The minimum is at $\theta^* = \arctan(-Q, P)$; both θ^* and $\theta^* + \pi$ are evaluated and the smaller d^2 is taken.

The top 10 closest-approach pairs are shown with connecting lines. When a pair is within $\Delta\theta \approx 0.15\,\text{rad}$ of their closest approach, the display flashes white.

11.4 Ghost Paths

In ghost mode, a selected particle's future trajectory is shown in three versions:

- **White solid**: actual kinematic orbit (integrating eqs. (6)–(7) for 40 substeps of $0.05\,\text{s}$).
- **Cyan dashed**: gas-only co-rotation (no settling, i.e. $v_{\rm t} = 0$).
- **Purple dashed**: free-fall vacuum trajectory (no drag, only gravity $g = 200\,\text{du}\,\text{s}^{-2}$ downward).

The comparison between the three paths makes the role of aerodynamic drag and the settling velocity physically intuitive.

11.5 v_t Distribution Display

Live kernel-density estimates are drawn for:

- **Yellow**: $v_{\rm t}$ distribution of all floating (non-levitated) particles.
- **Green**: $v_{\rm t}$ distribution of levitated particles.

The KDE uses Silverman's rule of thumb for bandwidth: $h = 1.06\,\hat{\sigma}\,n^{-1/5}$. An amber histogram shows $v_{\rm t}$ values of live aggregates.

All three are normalised independently (height = 1 at peak), so relative shapes are directly comparable regardless of population size.

11.6 Aggregate Size Histogram

A bar chart inside the left half of the drum shows the current aggregate population binned by monomer count in steps of 10 (bins 10–100), plus a gold bar for pebbles. The y-axis is fixed at a maximum of 10; count labels appear above bars when counts exceed the axis. The display updates continuously.

12 Collection Tray

The collection tray is a retractable metal blade that can be inserted into the drum to capture levitated particles, aggregates, and pebbles.

12.1 Arming

Pressing the **Collect** button ( ) arms the tray. The arm waits for the gold slot marker on the bolt ring to cross the 12-o'clock position on screen (canvas angle $-\pi/2$). At that moment, insertion begins. Pressing the button again while armed cancels arming.

12.2 Insertion Kinematics

The blade extends from the slot marker (the hinge point on the rim) inward along a fixed direction locked at fire time. The direction aims toward a point $(0.447R, 0.224R)$ in drum coordinates, producing a blade that sweeps a chord roughly $0.89R$ long.

Insertion sweeps 270 of drum rotation, divided into two phases:

1. **Cruise** (first 2/3): drum speed held constant at the firing value.
2. **Brake** (final 1/3): drum decelerates smoothly to zero via a cubic smoothstep ramp:

$$\omega(t) = \omega_0 \cdot \left(1 - 3u^2 + 2u^3\right), \quad u = (t - t_{\text{brake}})/\tau_{\text{brake}}. \tag{45}$$

Particles and aggregates within $2 \times \tau_{\text{thick}}$ of the tray line are captured and pinned to the upper blade surface, then co-rotate with the drum until insertion is complete.

13 Slice Widening (Doubling)

13.1 Physical Justification

The simulation models a single axial *slice* of the drum. Early in a run, this slice is injected with N_P particles. Through wall losses and aggregate/pebble formation, the active particle count decreases. Computational capacity is then underused.

The physical assumption is that adjacent axial slices have undergone the same process: they started with a similar particle distribution and have evolved identically (same levitation conditions, same statistics). Widening the simulated slice to include one adjacent slice should therefore approximately double the levitated population without introducing any particles that are physically implausible.

13.2 Implementation

Pressing duplicates all currently levitated particles and all currently levitated aggregates. Each copy is placed 90 ahead of the original in its orbit. The copy's position is computed as:

$$x' = x_c + r\cos(\phi + \pi/2) = x_c - r\sin\phi = x_c - y, \tag{46}$$

$$y' = r\sin(\phi + \pi/2) = r\cos\phi = x - x_c, \tag{47}$$

and the copy's velocity is set to the orbital velocity at the new position:

$$v'_x = -\omega y', \tag{48}$$
$$v'_y = +\omega x' - v_{\mathrm{t}}. \tag{49}$$

The 90 offset ensures no two copies are at the same position, avoiding initialisation artefacts. The copies inherit `inHighlightSince` from the originals so they count as immediately levitated. The button flashes without action if the drum is not spinning (orbits are undefined).

14 User Interface — Main Controls

14.1 Left Wing: Injection Parameters

Icon	Label	Description
	Particles	Number of particles per injection event $N_P \in \{1, 2, 3, 10, 30, 100, 300, 1000, 3000, 10000\}$.
	v_t **cm/s**	Mean terminal velocity $\bar{v}_{\mathrm{t}} \in \{2, 5, 10, 20, 30, 40, 50\}\,\mathrm{cm\,s^{-1}}$. Larger v_{t} produces visually larger particles. Overridden by the distribution panel when a custom profile is active.
	v_t **spread**	Relative Gaussian spread $s \in \{0, 0.01, 0.02, 0.05, 0.10, 0.20, 0.30, 0.50\}$. Must be $\geq$ 0.20 for aggregate formation and $\geq$ 0.30 for pebble formation.
	Δt **inject**	Injection time span $\Delta t_{\mathrm{inj}} \in \{0, 0.5, 1, ..., 10\}\,\mathrm{s}$. Particles are released uniformly over this interval.

Clicking the value display of VT or Spread when a custom distribution profile is active opens the distribution overlay directly, rather than cycling the value. The display shows the profile name in orange.

14.2 Right Wing: Action Controls

Icon	Label	Description
	Inject	Clears existing particles and releases a new batch. The mill start sound plays for the injection duration.
	Sound	Toggles all audio (motor hum, impact sounds, chimes). Default: on.
Ω	Ω	Locks the current drum rotation rate: decay is set to zero and no further acceleration is applied. Useful for sustained levitation experiments.

	Trails	Enables particle trail rendering. Each particle leaves a fading path showing its recent trajectory over 1 s. Disabled for $N_P \geq 300$ (performance).
	Lidar	Engages *lidar mode*: the drum darkens and particles are visible only when the scanning green laser beam crosses them. The laser makes one full revolution every 1.2 s (angular velocity $\omega_{\text{laser}} = 2\pi/1.2 \approx 5.24\,\text{rad}\,\text{s}^{-1}$).
	Microscope	Slides the ceiling-mounted viewport down to display microscope imagery or live rendered content, governed by the viewport rule system (§18).
	Chart	Slides the viewport down to display data charts or live rendered content.
	Lighting	Toggles between the dark laboratory background and the light daytime background.
	Game	Enters the 8-level progressive game mode (§16).
	Challenge	Enters timed challenge mode with a local leaderboard (§17).
	Info	Opens the user manual overlay.
	Reset	Clears all particles, aggregates, pebbles, and globes; resets the simulation to a clean state ready for a new injection.

15 Expert Panel

The expert panel is concealed behind a locked door at the bottom of the left wing. It can be opened by:

- Triple-tapping the title plate.
- Appending `?expert` to the URL.

The panel is divided into three sections: HUD, Processes, and System.

15.1 HUD — Visualisation Overlays

Icon	Name	Description
	Orbit centre projection	Draws the range of orbit centres $[v_t^{\min}/\omega, v_t^{\max}/\omega]$ as a horizontal dashed line on the drum equator, with dots at the endpoints and centre. Updates live as v_t, spread, or ω changes. Reveals the physical intuition behind eq. (8).
	Flash orbit	After each aggregate forms, briefly draws its orbit ring (cyan) alongside the mean-v_t orbit of the levitated population (amber), connected by a line showing the offset due to the Stokes kick.

	Encounter projections	Shows the closest-approach positions and separation distance for the top-10 aggregate pairs (§9.2). Flashes white when a pair is near approach.
	v_t distribution	Live KDE of levitated (green) and floating (yellow) particles, plus amber histogram of aggregate v_t values.
	Size histogram	Bar chart of current aggregate monomer count distribution, plus pebble count.
	Backdrop	Opaque backing sheet behind all overlays for contrast. Short press toggles; long press ($>$ 400 ms) opens an opacity slider.
◄	**Mode prev**	Step to previous analysis mode.
	Analysis HUD	Master power switch for the heatmap and overlay analysis engine. Activating resets all accumulators. Six modes cycle with ◄ ►: *vector field*, *ghost paths*, *orbit sample*, *density*, *dispersion*, *collision proxy*.
►	**Mode next**	Step to next analysis mode.

15.2 Processes

Icon	Name	Description
	Nanocoating	Prevents particle and aggregate loss at the wall. Particles are elastically reflected; aggregates are clamped and damped. Useful for accumulating large levitated populations.
	Sweep	Instantly removes all non-levitated floating particles. A single-use action (flashes, returns to passive).
	Widen slice	Duplicates all levitated particles and aggregates, placing copies 90 ahead in their orbits. Models widening the simulated drum slice.
	Stokes kick	When on (default off), aggregates receive a $\times 1.2$ v_t boost at formation and $\times 1.05$ per growth merge, reflecting higher Stokes number and cross-section shadowing.
	Aggregate formation	Green (default on): aggregates form normally from levitated particles. Red (off): the formation threshold is set to 10^6 (effectively disabled), useful for accumulating large levitated monomer populations without self-organisation.

	Growth mode	Three-state button: • *Green (Pebble)*: standard operation — 7 levitated aggregates merge into a pebble after the hold criterion. • *Amber (Grow)*: aggregate–aggregate collision growth active. Brownian perturbations, restoring spring, and pebble transition at $n = 100$. Pebble formation via the standard path is disabled. • *Red (Off)*: all growth beyond aggregate formation disabled.

15.3 System

Icon	Name	Description
	Distribution	Opens the injection profile panel. Three modes: Gaussian (uses wing sliders), bi-disperse (two groups with individual v_t, spread, ratio), and power-law (eq. (15)). Applies from next injection.
	Auto-omega	Three-state button: • *Off*: user controls ω manually. • *Green (Auto)*: live orbit-centring using eq. (15). • *Orange (Launch)*: waits for injection, computes optimal spin-up time, spins up, then tracks.
	Collect	Arms the collection tray. Waits for the slot marker to reach 12 o'clock, then inserts the blade and stops the drum.
◀	**Speed down**	Reduce simulation speed one gear.
	Speed	Shows current speed gear; press to reset to 1×.
▶	**Speed up**	Increase simulation speed one gear.
	Strobe	Redraws once per drum revolution (stroboscopic mode).
	Slow motion	Short press: 5× slow during merge animations. Long press ($> 2\,\mathrm{s}$): activates *Fast Chain* mode (dramatically reduced thresholds for rapid prototyping).
	Zoom	Zooms the view so the levitation zone fills the drum area.

16 Game Mode

Game mode presents 8 progressive levels. The game locks the injection parameter selectors during play and monitors simulation state each frame.

#	Name	Parameters / Goal / Fail

1	First Contact	$N_P = 3$, $v_t = 20$, $s = 0$ *Goal:* ≥ 1 levitated for 3 s *Fail:* all particles lost
2	Finding the Sweet Spot	$N_P = 3$, $v_t = 30$, $s = 0$, trails *Goal:* ≥ 2 levitated for 5 s *Fail:* fewer than 2 floating
3	Reading the Orbits	$N_P = 5$, $v_t = 50$, $s = 0$, trails *Goal:* ≥ 2 levitated for 5 s *Fail:* fewer than 2 floating
4	A Spread of Sizes	$N_P = 15$, $v_t = 30$, $s = 0.15$, trails *Goal:* ≥ 4 levitated for 5 s *Fail:* fewer than 4 floating
5	Flying Blind	$N_P = 20$, $v_t = 30$, $s = 0.15$, lidar *Goal:* ≥ 6 levitated for 8 s *Fail:* fewer than 6 floating
6	Crowded Skies	$N_P = 50$, $v_t = 30$, $s = 0.15$ *Goal:* ≥ 10 levitated for 10 s *Fail:* fewer than 10 floating
7	Encouraging Collisions	$N_P = 100$, $v_t = 30$, $s = 0.20$ *Goal:* ≥ 1 aggregate formed *Fail:* timeout after 40 revolutions
8	More Energetic Encounters	$N_P = 300$, $v_t = 30$, $s = 0.30$, trails *Goal:* ≥ 1 pebble formed *Fail:* timeout after 100 revolutions

Structure-forming levels (7–8) use a wall-clock-scaled timeout: $t_{\max} = n_{\text{revs}} \times 3\,\text{s}$ (assuming ≥ 1 rev per 3 s). A win confirmation delay of 5 s is applied before the success sheet appears, giving the player time to observe the formed structure.

Completing level 8 shows a transition sheet to the free-play sandbox, with the level-8 parameters (300 particles, 30% spread) pre-applied.

17 Challenge Mode

Challenge mode runs a timed levitation contest with a weighted score:

$$\text{Score} = N_{\text{part}} + 10\,N_{\text{agg}} + 100\,N_{\text{pebble}} + 1000\,N_{\text{planet}}, \tag{50}$$

where each count is taken at run end. A particle counts if it has been continuously levitated for ≥ 3 full revolutions.

The run ends when all injected particles are either levitated, stuck, or lost, or when the time limit $\Delta t_{\text{limit}} = 60 + \Delta t_{\text{inj}}$ seconds is exceeded.

Default challenge parameters: $N_P = 300$, $v_t = 30\,\text{cm}\,\text{s}^{-1}$, $s = 0.30$, $\Delta t_{\text{inj}} = 3\,\text{s}$. All can be overridden by URL parameters (see §20).

Scores are stored in browser `localStorage` under the key `levitation_highscores` as a JSON array of {`name, score, np, date`} objects. The top 7 are displayed.

18 Viewport System

The viewport is a brass-framed screen that slides down from above the drum on the Microscope or Chart button press. It displays either static image or video assets, or live canvas-rendered content, driven by a rule engine.

18.1 Rule Evaluation

Every 1/5 s, the engine evaluates a bank of rules for the active channel (`microscope` or `chart`). Each rule has:

- A `when` predicate receiving a `ctx` snapshot of simulation state (particle counts, drum state, mode flags, etc.).
- An `asset` (filename string, array for random selection, or a `{type, src/name}` descriptor).
- A `priority` (higher wins; default 0).

The highest-priority matching rule is selected. Asset changes are crossfaded over 0.4 s with a 1.0 s debounce to prevent flickering.

18.2 Live Renderers

Any asset of type `render` calls a registered JavaScript function that paints into the viewport's off-screen canvas each frame:

```
registerViewportRenderer('myPlot', function(g, w, h, ctx, t) {
    // g: 2D context  w,h: pixel size  ctx: simulation snapshot
       t: sim time
});
```

The `ctx` object exposes particle counts, drum state, mode flags, and current parameters, enabling data-driven visualisations.

18.3 Context Object Summary

Key fields available in `ctx`:

- `ctx.particles.{floating, levitated, lost, total, ... }`
- `ctx.aggregates.{active, levitated, totalFormed}`
- `ctx.pebbles.{count}`
- `ctx.drum.{omega, period, spinning, direction}`
- `ctx.params.{N_P, V_T, VT_SPREAD, DT_INJECT}`
- `ctx.mode.{game, gameLevel, challenge, running}`
- `ctx.flags.{laserOn, trailsOn, theme}`

19 Audio System

All audio is synthesised in real time using the Web Audio API.

Sound	Description
Motor hum	Two sawtooth oscillators at 60 Hz and 60.6 Hz (slight detuning for beating effect) through a 400 Hz low-pass filter, plus bandpass-filtered noise at 300 Hz. Gain and pitch track $\lvert\omega\rvert$ smoothly with time constant $\tau = 0.06$ s.
Tink	Short metallic impact: $220 \rightarrow 90$ Hz sine sweep with 500 ms cooldown. Fires on particle–wall collision when $N_P \leq 30$.
Snap	9 accelerating clicks (sawtooth transients at 2500–4300 Hz, inter-click gap multiplied by 0.85 each step). Fires on aggregate formation.
Aggregate merge	Soft sine thud descending from 600 to 220 Hz as aggregate size increases from 20 to 100 monomers. Fires on growth merges.
Crunch	500 random sawtooth bursts (2000–5000 Hz, 80 ms random offsets, 0.8 s total). Fires on pebble formation.
Chime	Two-note (880 and 1320 Hz) sine arpeggio with 150 ms cooldown. Fires when a particle achieves levitation.
Golden chime	Four-note arpeggio (C5, E5, G5, D6) with decaying envelope. Fires when the aggregate gauge or pebble gauge first appears.
Golden thud	Sine sweep $160 \rightarrow 70$ Hz, amplitude proportional to collision speed. Fires on strong golden-ball wall or ball–ball impacts.
Mill start	Full five-component sequence (blade, hiss, grind, thud, body) lasting the injection duration. Fires on Inject press.
Expert door	Heavy metallic clunk plus a 3-second cavernous low-frequency reverb tail. Fires when the expert door is clicked while closed.

The `AudioContext` is created and suspended on first load. It is resumed on the first user gesture (`pointerdown` or `keydown`), as required by browser autoplay policies. Muting sets the master gain to 0 with a 50 ms ramp; the audio graph remains active. Tab visibility changes also mute and unmute smoothly.

20 URL Parameters

Parameters are parsed from `window.location.search` at load time.

Parameter	Effect
`?expert`	Opens the expert panel and activates the Ω lock.
`?game`	Starts Game Mode at level 1, splash suppressed.
`?game=N`	Starts Game Mode at level N (1-indexed; out-of-range falls back to 1).
`?challenge`	Starts Challenge Mode with default challenge parameters, splash suppressed.
`?challenge&np=N`	Override particle count in challenge (also `vt`, `spread`, `dt`).

`?modern`	Applies the clinical lab visual theme (light gray metals, teal buttons).
`?pfeiffer`	Applies the Pfeiffer variant (red structural elements, modern theme).
`?verify`	Developer smoke-test mode: $N_P = 3000$, $s = 0.30$, Ω lock on.

Multiple parameters can be combined: e.g. `?expert&game=3` or `?modern&challenge&np=1000`.

21 Performance Notes

- **Trail rendering** is disabled for $N_P \geq 300$ (the `maxN` tuning constant). At high particle counts, per-particle trail segment storage and rendering becomes the bottleneck.
- **Particle glow** (radial gradient per particle) is skipped for $N_P \geq 300$; only the core circle is drawn.
- **Injection puffs** are limited to one burst of 5 puffs (one per nozzle) at injection start, regardless of N_P. Each puff generates three `createRadialGradient` calls per frame for its 0.42 s lifetime. At $N_P = 3000$ over a 2 s injection window, the original per-particle puff design created up to $\sim 1500\,\mathrm{puffs\,s^{-1}}$, causing significant frame-rate drops on mobile WebKit (observed on iPad).
- **Encounter prediction** updates its cache at $\approx 5\,\mathrm{Hz}$ (every 12 frames at 60 fps) to keep the $O(n^2)$ pairwise computation from dominating each frame.
- **Heatmap** finalisation runs once per full drum revolution, not every frame; the per-frame cost is only the accumulation of one position per particle per substep.

A Tuning Constants Reference

All tuning constants are defined in `config.js` under the `TUNING` object and backed up in `TUNING_DEFAULT` for restoration after overrides.

Key	Default	Description
Drum		
`drum.slipRate`	0.7	Omega tracking lag (λ_s, $\mathrm{s^{-1}}$)
`drum.omegaDecay`	1/30	Omega target decay (λ_d, $\mathrm{s^{-1}}$)
`drum.swipeGain`	0.003	Swipe to omega conversion
`drum.omegaMax`	3π	Max angular velocity ($\mathrm{rad\,s^{-1}}$)
Highlight zone		
`highlight.cx`	50	Zone centre x (du)
`highlight.radius`	50	Zone radius (du)
Particles		
`particle.settleFloor`	2.0	Minimum v_t (cm/s)
`particle.sizeRefVt`	30	Reference v_t for size scaling
`particle.collisionR`	0.5	Collision radius (du)

Aggregates		
`aggregate.minLevitated`	30	Min levitated for formation
`aggregate.mergeCount`	10	Monomers per aggregate
`aggregate.initialHoldRevs`	3	Hold revs before first aggregate
`aggregate.subseqHoldRevs`	1	Hold revs for subsequent aggregates
`aggregate.sizeMult`	10	Visual size multiplier
`aggregate.mergeDur`	0.6	Merge animation duration (s)
`aggregate.minSpread`	0.20	Min spread for formation
`aggregate.vtFactor`	1.2	Stokes kick factor
`aggregate.vtGrowFactor`	1.05	Per-growth-merge kick factor
`aggregate.brownian`	2.0	Brownian kick amplitude ($\mathrm{cm\,s^{-1}}$)
`aggregate.restoreK`	0.5	Restoring spring constant ($\mathrm{s^{-1}}$)
Pebbles (egg)		
`egg.nCrit`	7	Aggregates required for pebble merge
`egg.holdTarget`	3	Hold revs for first pebble
`egg.holdSubseq`	3	Hold revs for subsequent pebbles
`egg.maxBalls`	30	Maximum simultaneous pebbles
`egg.minSpread`	0.30	Min spread for pebble formation
Golden balls		
`ball.gravity`	200	Gravitational acceleration ($\mathrm{du\,s^{-2}}$)
`ball.radius`	2.5	Physical radius (du)
`ball.wallE`	0.85	Wall restitution coefficient
`ball.wallFriction`	0.92	Wall tangential retention
`ball.ballE`	0.90	Ball–ball restitution coefficient
Globes		
`globe.nCrit`	3	Pebbles required per globe
`globe.radius`	12.5	Globe physical radius (du)
`globe.mergeDur`	1.2	Merge animation duration (s)
`globe.repulsion`	15000	Globe–globe repulsion constant
`globe.limit`	7	Maximum simultaneous globes
Solar system		
`solar.inclCos`	0.3	$\cos i$ of orbital plane tilt
`solar.omegaBase`	3.006	Keplerian base rate ($\mathrm{rad\,s^{-1}}$)
`solar.maxR`	80	Max orbit radius (du)
`solar.persp`	0.10	Perspective size variation
Audio		
`audio.masterGain`	0.6	Master volume (0–1)
`audio.millBlade`	0.085	Mill blade component level
`audio.millGrind`	0.85	Mill grind component level

B Button Icon SVG Sources

The following SVG fragments are the source code for each custom button icon. They should be saved as individual files in an `icons/` subdirectory for inclusion via `\btnicon{}` in this document. All icons use a 24×24 viewBox and `currentColor` for strokes and fills that should respond to button state; hardcoded colours are used for elements that should remain constant across states.

B.1 Widen Slice (`btnProcDouble`)

```
<svg viewBox="0␣0␣24␣24" width="20" height="20">
  <!-- Shadow copy (right): no left side -->
  <path d="M␣11␣4␣L␣16␣4␣L␣16␣20␣L␣11␣20"
        fill="none" stroke="#8a6040" stroke-width="1.5"/>
  <circle cx="12.8" cy="6.5"  r="0.7" fill="#8a6040"/>
  <circle cx="14.2" cy="9.5"  r="0.7" fill="#8a6040"/>
  <circle cx="13.5" cy="12.5" r="0.7" fill="#8a6040"/>
  <circle cx="12.8" cy="15.5" r="0.7" fill="#8a6040"/>
  <circle cx="14.0" cy="18.0" r="0.7" fill="#8a6040"/>
  <!-- Original (left) -->
  <rect x="6" y="4" width="5" height="16"
        fill="none" stroke="currentColor" stroke-width="1.5"/>
  <circle cx="7.8"  cy="6.5"  r="0.7" fill="currentColor"/>
  <circle cx="9.2"  cy="9.5"  r="0.7" fill="currentColor"/>
  <circle cx="7.6"  cy="12.5" r="0.7" fill="currentColor"/>
  <circle cx="9.0"  cy="15.5" r="0.7" fill="currentColor"/>
  <circle cx="8.0"  cy="18.0" r="0.7" fill="currentColor"/>
</svg>
```

The left rectangle uses `currentColor` so it responds to the button's amber base, hover, and ON-state colours. The right rectangle and its particles are hardcoded to `#8a6040` to remain consistently dim regardless of button state.